PPTrans

PowerPoint translation with verified package integrity

V2 PUBLIC SOURCE / PRE-RELEASE

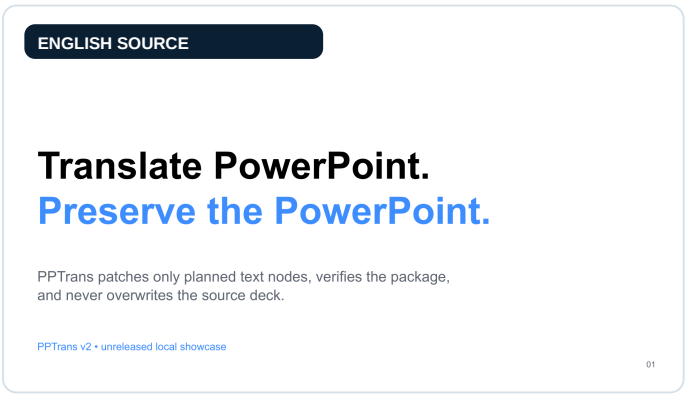
HEHAN ZHAO | ENGINEERING CASE STUDY | EVIDENCE SNAPSHOT 2026-08-31

DEFENSIVE OOXML | STRICT AI CONTRACTS | TRANSACTIONAL OUTPUT

Change the intended text nodes - and prove everything else stayed within plan.

PPTrans turns translation into a source-bound transaction across untrusted OOXML and AI output.

ENGLISH SOURCE



CURATED ZH-CN



Real changed-text fixture through the exact-ID patch and verification path. Native LibreOffice 26.8.0.3 renders; this is not provider-quality evidence.

1 DEFEND ZIP + XML | 2 PLAN STABLE IDS | 3 VALIDATE AI OUTPUT | 4 PATCH STAGED COPY | 5 VERIFY ALL CHANGES | 6 PUBLISH ATOMICALLY

WHAT I ENGINEERED

- Led v2 product direction and the end-to-end re-architecture around immutable **DeckPlan** and **PatchSet** contracts.
- Built defensive ZIP/XML intake, source-hash binding, stable text addresses, and explicit support and resource limits.
- Made OpenAI and Anthropic adapters fail closed on partial, reordered, duplicated, invented, or schema-invalid IDs.
- Added staged patching, planned-and-untouched text verification, package checks, and atomic no-clobber publication.

NO-SPEND PROVIDER PREVIEW

41 units in **2 logical calls** | 2,799 source/context chars | 8,867 serialized request chars | largest 5,903

Zero-memory-hit upper bounds. The preview loads no credential, provider SDK, translation memory, or output path and makes no provider/API request. Counts are not tokens, price, latency, availability, or quality estimates.

MEASURED EVIDENCE

541 PASSING TESTS	**92.12%** BRANCH-AWARE COVERAGE
9,346 GENERATED EXAMPLES	**3 / 41 / 45** SLIDES / UNITS / SPANS
3 CHANGED SLIDE XML PARTS	**58.770 / 60.972 ms** MEDIAN / P95 LOCAL CORE

EVIDENCE BOUNDARY

- The synthetic, author-reviewed zh-CN deck proves changed text and package preservation, not production-provider translation quality.
- The benchmark covers the deterministic local core only; it excludes provider, network, memory, rendering, cost, and translation-quality time.
- LibreOffice evidence does not claim Microsoft PowerPoint pixel identity, automatic layout fit, universal compatibility, or maximum practical deck size.

STATUS AND PROVENANCE

PPTrans v2 is a **public-source pre-release engineering showcase**. It is a substantial re-architecture with repository history and upstream attribution retained; it is not presented as clean-room work or as unrelated to its predecessor. No stable package release has been published yet.

Configured Python support: 3.11-3.13 | Skills: OOXML/XML | typed domain design | schema-bound LLM adapters | transactional I/O | property testing | CI and security

Source: github.com/Z-MarkUs/PPTrans | Demo: examples/pptrans-demo.en.pptx -> examples/pptrans-demo.zh-CN.pptx
Evidence basis: 63a0e95 | local-test audit JSON | exact-rebuild QA JSON | raw benchmark JSON | CI matrix | NOTICE.md